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Data-Driven Prescriptive Planning for the RRS Logistics Network Under Demand Uncertainty

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Abstract

This study develops a data-driven framework for logistics planning under demand uncertainty by integrating predictive modeling with prescriptive optimization. Motivated by real-world multi-echelon distribution systems, such as the RiRiShun logistics network, we consider a three-level network consisting of central distribution centers (CDCs), regional distribution centers (RDCs), and destination nodes, where decisions involve routing flows and allocating capacity under uncertain demand. We compare four approaches: Expected Value (EV), Sample Average Approximation (SAA), Predict-Then-Optimize (PTO), and Predictive–Prescriptive (PP) optimization. Demand uncertainty is modeled using machine learning methods, including k-nearest neighbors (kNN) and decision trees, which are integrated into downstream optimization decisions. The results demonstrate that deterministic approaches, such as EV, perform poorly because they cannot capture demand variability. SAA provides limited improvement by incorporating sampled scenarios, while PTO significantly reduces cost by leveraging demand predictions. The proposed PP framework achieves the best performance by directly embedding predictive information into the optimization model, resulting in lower total cost and reduced shortage. Furthermore, the analysis highlights the importance of incorporating uncertainty into logistics decision-making and shows that data-driven optimization approaches can substantially improve both operational efficiency and service reliability.
Keywords: logistics optimization, demand uncertainty, predictive–prescriptive analytics, sample average approximation, machine learning, network flow, supply chain analytics, k-nearest neighbors, decision trees

1 Introduction

1.1 Problem Definition

Modern logistics networks operate under persistent and substantial demand uncertainty. For large-scale retail and e-commerce fulfillment systems, such as the RRS (Rapid Retail and Supply) network, determining optimal shipment volumes and routing decisions before demand is realized is a central operational challenge. Incorrect or suboptimal decisions can result in either excessive transportation costs due to over-allocation of capacity or shortage costs and service failures when demand is underestimated.

This project develops a data-driven prescriptive planning framework for the RRS logistics network, which is organized as a three-level aggregated service network inspired by real-world logistics systems such as the RiRiShun network Guo et al. (2024). The network consists of:

- **Origin Centers (CDC)** — supply nodes from which goods are dispatched,
- **Intermediate Transshipment Centers (RDC)** — regional consolidation hubs,
- **Demand Nodes** — aggregated last-mile transfer centers (LTC) and customer zones (CZ).

The core decision problem is to determine optimal flow volumes across this network for each load class and planning period, subject to supply limits, transshipment capacity, service-time feasibility, and penalties for unmet demand. The planning horizon is defined at a weekly level to align with tactical capacity allocation decisions.

A key complexity arises from the fact that future demand at each destination node is unknown when decisions are made. Therefore, demand must be represented using appropriate uncertainty modeling techniques. To address this, we evaluate and compare four decision-making frameworks:

- Expected Value (EV),
- Sample Average Approximation (SAA),
- Predict-Then-Optimize (PTO),
- Predictive–Prescriptive (PP) optimization using machine learning models such as k-Nearest Neighbors (kNN) and Decision Trees.

1.2 Key Research Questions

This study is guided by four interconnected research questions. First, we investigate how different representations of demand uncertainty—ranging from simple historical averages and scenario sampling to point predictions and data-driven weighting—affect transportation cost, shortage cost, and overall service performance within the RRS network.

Second, we examine whether machine learning models, such as kNN and Decision Tree regressors, can improve planning quality relative to traditional statistical approaches, such as SAA. In particular, we explore whether these models can generate more informative demand scenarios or assign contextually relevant weights to historical observations.

Third, we analyze the impact of network structure on decision outcomes by examining how the maximum allowable delivery time threshold ($T_{\max}$) limits feasible arcs and affects total cost. This provides insight into the trade-off between service-level constraints and operational efficiency.

Finally, we consider computational tractability and scalability by evaluating whether the three-level aggregated transshipment model serves as a reliable prototype for large-scale logistics planning. We also discuss how this structure can be extended to represent more detailed and realistic network configurations.

2 Data Exploration

2.1 Overview of Datasets

The modeling pipeline utilizes six datasets from the RRS data repository, each providing complementary information on orders, shipment characteristics, and network structure. Together, these datasets enable a comprehensive representation of logistics operations, consistent with real-world systems such as the RiRiShun network Guo et al. (2024).

- **Orders.csv (213 rows, 12 columns):** The primary transactional dataset, where each row represents a shipment order. Variables include origin and destination center codes, order date, delivery method, distance information, and order amount. After removing trivial same-origin-destination records, 213 valid orders remain, covering 25 origin centers and 67 destination centers.
- **SKU_details.csv (673 rows, 11 columns):** Contains item-level attributes such as material code, quantity, volume, weight, and height. These features are aggregated to the order level to construct load classes. Coverage is complete across all orders.
- **Appointment_details.csv (500 rows, 7 columns):** Includes appointment scheduling and aging classifications. This dataset is used for optional feature engineering and service-level analysis.
- **Delivery_details.csv (3,951 rows, 7 columns):** Captures node-level shipment paths, allowing reconstruction of observed routes and inference of feasible network arcs.
- **Client_details.csv (149 rows, 2 columns):** Provides client type classifications for potential demand segmentation.
- **Distance_information.csv (103 $\times$ 104 matrix):** Contains pairwise distances between network nodes. This dataset serves as the basis for transportation cost estimation and travel-time approximation.

2.2 Key Descriptive Findings

An exploratory analysis of the orders and SKU datasets reveals significant variability in shipment sizes. Order volumes range from approximately 63,920 to over 1,773,000 cubic units, with a standard deviation comparable to the mean. To capture this heterogeneity, shipments are segmented into three load classes: small, medium, and

large.
Weight distributions exhibit a narrower range, with most orders between 25 and 60 kilograms, further supporting the three-class segmentation.
On the temporal dimension, orders are aggregated into weekly periods. This aggregation smooths short-term fluctuations while preserving variability relevant for tactical planning decisions.
The distance matrix spans 103 centers. Applying a service-time constraint with a maximum delivery time of 96 hours and an assumed travel speed of 60 km/h limits feasible arcs to approximately 5,760 km. This significantly reduces the network size and improves computational tractability.
Overall, the dataset is of high quality, with complete SKU coverage, no missing key variables, and no duplicate records.

2.3 Modeling Challenges

Several challenges arise in constructing the optimization model from the raw datasets:

- **Network size and complexity:** The full dataset includes numerous centers and potential arcs, leading to computational challenges. To address this, the network is reduced to a smaller prototype with 16 origin centers and 58 demand nodes.
- **Demand aggregation:** Raw shipment data must be aggregated to the (destination, load class, week) level. Load class boundaries are defined using quantiles of volume and weight.
- **Supply estimation:** Direct data on inventory and capacity is unavailable. Supply is approximated using historical shipment counts by origin, introducing additional uncertainty.
- **Arc feasibility:** Service-time constraints are enforced using distance-based travel time approximations. This simplification does not account for real-world routing variability.
- **Scenario generation:** Stochastic methods such as SAA require multiple demand scenarios. Due to limited data, the number of scenarios is restricted, which may affect solution robustness.

3 Methodology

Network Construction

The logistics network is modeled as a three-level aggregated transshipment system derived from a more detailed multi-echelon structure. In real-world logistics systems, such as the RRS network, shipments typically move through multiple layers, including central distribution centers (CDCs), regional distribution centers (RDCs), local transfer centers (LTCs), and last-mile delivery nodes. Directly modeling all four levels leads to a large-scale network with significant computational complexity.
To address this challenge, we adopt an aggregation strategy that consolidates the lower-level delivery structure (LTCs and customer zones) into a unified set of demand nodes. As a result, the network is represented as a three-level system consisting of:

- **Origin Centers (CDC):** supply nodes from which shipments are dispatched,
- **Intermediate Transshipment Centers (RDC):** regional consolidation hubs,
- **Demand Nodes:** aggregated representation of last-mile transfer centers and customer zones.

This aggregation significantly reduces network size while preserving the essential flow structure and routing flexibility required for decision-making. It enables the model to remain computationally tractable while still capturing the key operational characteristics of the logistics system.

Origin centers and demand nodes are selected based on historical shipment activity. Specifically, up to 20 origin centers and 15 destination nodes are retained based on configurable thresholds. After role separation and filtering, the resulting network consists of 16 exclusive CDCs, 9 RDCs, and 58 demand nodes. Intermediate transshipment centers are identified as nodes that appear both as origins and destinations in the historical order data.
Candidate arcs are generated and filtered according to two key criteria.
First: *service feasibility*: a direct arc from origin i to demand node j is included only if the associated travel time does not exceed the maximum allowable delivery time $T_{\max}$. Similarly, a transshipment route through an intermediate center m is included only if the combined travel time across both legs satisfies this constraint.
Second: *observed connectivity*: arcs are further restricted to historically observed routes or those deemed plausible based on distance thresholds. This ensures that the network reflects realistic routing patterns and avoids infeasible or rarely used connections.
Finally, capacity constraints at intermediate centers are approximated using historical shipment data. Specifically, the throughput capacity of each RDC is estimated by scaling observed order volumes by 1.25, providing a conservative buffer to account for demand variability and operational uncertainty.

4 Optimization Model

The core optimization model is formulated as a multi-commodity network flow problem and implemented in Pyomo. The objective is to minimize total transportation cost and shortage penalties across load classes and time periods. Under stochastic settings, shortage costs are evaluated across demand scenarios.
Transportation cost on each arc is computed as a unit cost rate multiplied by distance, while unmet demand is penalized to capture service failures. The model supports both direct shipments and transshipment flows through intermediate centers.

Modeling Assumptions

To ensure tractability and alignment with available data, the following assumptions are adopted:

- Demand is uncertain and represented through scenarios or predictive models depending on the solution approach.
- Supply at origin centers is approximated using historical shipment volumes.
- Transportation cost is linear in distance and flow volume.
- Travel time is proportional to distance, assuming a constant speed of 60 km/h.
- A shipment is feasible only if it satisfies a maximum delivery time constraint $T_{\max}$.
- Intermediate centers have limited throughput capacity.
- Unmet demand is allowed and penalized in the objective function.
- The planning horizon is discretized into weekly periods.

Network Structure

The logistics network is modeled as a three-level aggregated transshipment system:

$$\text{Origin Centers (CDC)} \rightarrow \text{Intermediate Centers (RDC)} \rightarrow \text{Demand Nodes (LTC + CZ)}$$

In addition, direct shipments from origin centers to demand nodes are allowed when feasible.

Sets and Indices

- $i \in I$: set of origin centers (CDC)

- $m \in M$: set of intermediate centers (RDC)
- $j \in J$: set of demand nodes
- $l \in L$: set of load classes (Small, Medium, Large)
- $t \in T$: set of time periods (weeks)
- $s \in S$: set of demand scenarios

Parameters

- S_{ilt}: available supply at origin i, load class l, time t
- D^s_{jlt}: demand at node j, load class l, time t, scenario s
- c_{im}, c_{mj}, c_{ij}: unit transportation cost on arcs
- R_{mt}: capacity of intermediate center m at time t
- p_{jl}: penalty cost per unit of unmet demand
- $\tau_{im}, \tau_{mj}, \tau_{ij}$: travel time on arcs
- $T_{\max}$: maximum allowable delivery time
- $a_{im}, b_{mj}, d_{ij} \in \{0, 1\}$: arc feasibility indicators

Decision Variables

- $x_{imlt} \geq 0$: flow from origin i to intermediate center m
- $y_{mjlt} \geq 0$: flow from intermediate center m to demand node j
- $z_{ijlt} \geq 0$: direct flow from origin i to demand node j
- $u^s_{jlt} \geq 0$: unmet demand at node j under scenario s

Arc Feasibility Constraints

Arcs are restricted based on service-time feasibility. Travel time is approximated using distance.
Direct shipment is allowed only if:

$$\tau_{ij} \leq T_{\max}$$

Transshipment is allowed only if:

$$\tau_{im} + \tau_{mj} \leq T_{\max}$$

These conditions define feasibility indicators a_{im}, b_{mj}, and d_{ij}, which restrict allowable flows in the network.

Objective Function

The objective is to minimize total cost, including transportation and shortage penalties:

$$\min \sum_{i,m,l,t} c_{im} x_{imlt} + \sum_{m,j,l,t} c_{mj} y_{mjlt} + \sum_{i,j,l,t} c_{ij} z_{ijlt} + \frac{1}{|S|} \sum_{s \in S} \sum_{j,l,t} p_{jl} u^s_{jlt}$$

Constraints

Supply constraints:

$$\sum_m x_{imlt} + \sum_j z_{ijlt} \leq S_{ilt} \quad \forall i, l, t$$

Flow conservation at intermediate centers:

$$\sum_i x_{imlt} = \sum_j y_{mjlt} \quad \forall m, l, t$$

Capacity constraints at RDC:

$$\sum_{i,l} x_{imlt} \leq R_{mt} \quad \forall m, t$$

Demand satisfaction:

$$\sum_m y_{mjlt} + \sum_i z_{ijlt} + u^s_{jlt} = D^s_{jlt} \quad \forall j, l, t, s$$

Non-negativity:

$$x_{imlt}, y_{mjlt}, z_{ijlt}, u^s_{jlt} \geq 0$$

Model Interpretation

The objective function minimizes the total system cost, consisting of two components: transportation costs and shortage penalties. Transportation costs are incurred for all flows along network arcs and are proportional to both distance and shipment volume. The shortage penalty term captures the cost of unmet demand and is averaged across scenarios to account for demand-forecast uncertainty.

The supply constraints ensure that the total outbound flow from each origin center does not exceed the available supply for each load class and time period. The flow conservation constraints at intermediate centers enforce that all incoming flow to an RDC is fully redistributed to downstream nodes, ensuring consistency in transshipment operations.

Capacity constraints at intermediate centers limit the total throughput that each RDC can handle in a given time period, reflecting operational limitations in handling and processing shipments.

The demand satisfaction constraints ensure that all realized demand is either fulfilled through incoming shipments or accounted for as unmet demand. The inclusion of the unmet demand variable allows the model to remain feasible even when supply or network constraints prevent full demand fulfillment.

Finally, the arc feasibility conditions restrict flows to only those routes that satisfy service-level requirements, ensuring that all shipments meet the maximum allowable delivery time. Together, these constraints ensure that the solution is both operationally feasible and aligned with service quality expectations.

5 Optimization Approaches

We evaluate four approaches to representing demand uncertainty within the optimization framework. Each approach differs in how demand is modeled and incorporated into the decision-making process.

Expected Value (EV)

In the Expected Value approach, demand uncertainty is ignored by replacing stochastic demand with its historical mean. For each (j, l, t):

$$\hat{D}_{jlt} = \mathbb{E}[D_{jlt}]$$

The optimization problem becomes deterministic:

$$\min \sum_{i,m,l,t} c_{im} x_{imlt} + \sum_{m,j,l,t} c_{mj} y_{mjlt} + \sum_{i,j,l,t} c_{ij} z_{ijlt} + \sum_{j,l,t} p_{jl} u_{jlt}$$

subject to:

$$\sum_m y_{mjlt} + \sum_i z_{ijlt} + u_{jlt} = \hat{D}_{jlt} \quad \forall j, l, t$$

This approach provides a baseline solution but ignores demand variability, which may lead to suboptimal performance under uncertainty.

Sample Average Approximation (SAA)

The SAA approach models uncertainty through a finite set of demand scenarios S, generated via bootstrap sampling from historical data. The objective minimizes the expected cost across scenarios:

$$\min \frac{1}{|S|} \sum_{s \in S} \left[\sum_{i,m,l,t} c_{im} x_{imlt} + \sum_{m,j,l,t} c_{mj} y_{mjlt} + \sum_{i,j,l,t} c_{ij} z_{ijlt} + \sum_{j,l,t} p_{jl} u^s_{jlt} \right]$$

Flow decisions (x, y, z) are shared across scenarios (here-and-now decisions), while unmet demand u^s_{jlt} is scenario-dependent (wait-and-see).
This approach provides a stochastic solution with improved robustness compared to EV.

Predict-Then-Optimize (PTO)

In the PTO framework, demand is first predicted using machine learning models, including k-Nearest Neighbors (kNN) and Decision Trees. The predicted demand:

$$\hat{D}_{jlt}$$

is then used as deterministic input to the optimization model:

$$D_{jlt} \leftarrow \hat{D}_{jlt}$$

The optimization problem retains the same structure as the EV formulation. This approach captures temporal patterns and trends but treats predictions as certain, ignoring residual uncertainty.

Predictive–Prescriptive Optimization (PP)

The Predictive–Prescriptive approach integrates machine learning and optimization by using predictive models to generate scenario weights rather than point forecasts. Let $S(x)$ denote a set of scenarios conditioned on feature vector x. The objective becomes:

$$\min \sum_{s \in S(x)} w^s \cdot \text{cost}(s)$$

where w^s represents the importance weight of scenario s.
For kNN:

$$w^s = \begin{cases} \frac{1}{k}, & \text{if } s \text{ is among the } k \text{ nearest neighbors} \\ 0, & \text{otherwise} \end{cases}$$

For Decision Trees:

$$w^s = \begin{cases} \frac{1}{|\text{leaf}(x)|}, & \text{if } s \text{ belongs to the same leaf} \\ 0, & \text{otherwise} \end{cases}$$

This approach allows the optimization to emphasize historically relevant demand patterns, effectively combining prediction and decision-making into a unified framework.

5.1 Load Class Construction

Products are segmented into three load classes (small, medium, large) based on order-level volume and weight quantiles derived from SKU-level aggregation.
For each order, total volume and weight are computed by aggregating SKU-level attributes. Quantile thresholds are then applied to assign each order to a load class. This classification preserves the heterogeneity of shipment sizes and enables more accurate modeling of transportation cost and capacity utilization.
The resulting load classes are consistently used across the demand aggregation, optimization, and evaluation stages.

5.2 Evaluation Framework

All approaches are evaluated using a held-out test period following the train-test split date of June 20, 2019. Performance is assessed based on out-of-sample results.
The following metrics are used:

- Total transportation cost
- Total shortage cost
- Total cost (sum of transportation and shortage cost)
- Service rate (fraction of demand satisfied)

Evaluation is conducted under identical realized-demand conditions to ensure a fair comparison across models. In addition, the impact of service-level constraints can be analyzed by varying the maximum allowable delivery time $T_{\max}$. Tighter thresholds reduce the set of feasible arcs, potentially increasing transportation costs, while more relaxed thresholds improve routing flexibility.

6 Results and Evaluation

6.1 Evaluation Objectives and Research Questions

The evaluation is designed to address the key research questions outlined in the introduction. In particular, we assess:

- How different representations of demand uncertainty impact cost and service performance,
- Whether machine learning-based approaches improve decision quality,
- How the integration of prediction and optimization affects robustness,
- Whether the proposed framework remains computationally tractable at scale.

All models are evaluated on a held-out test set using realized demand, ensuring a fair out-of-sample comparison.

6.2 Parameter Settings and Experimental Design

The optimization model is solved under consistent parameter settings across all approaches.

- Transportation cost rate: 0.015 per unit-distance
- Shortage penalty: 100 per unit unmet demand

- Maximum delivery time: $T_{\max} = 96$ hours
- Planning horizon: weekly aggregation

For machine learning models:

- kNN: $k = 5$
- Decision Tree: maximum depth = 5

Scenario-based models (SAA and PP) use a limited number of demand scenarios due to computational constraints, reflecting realistic large-scale deployment considerations.

6.3 Scalability Considerations

A key requirement of the modeling framework is scalability to large logistics networks. The original dataset represents a multi-level network with a large number of nodes and potential arcs. Solving the full problem directly with multiple demand scenarios would be computationally intractable.
To address this, three key design choices are implemented:

- **Network aggregation:** The original multi-echelon structure is reduced to a three-level network, significantly reducing the number of decision variables.
- **Arc filtering:** Feasible arcs are restricted using service-time constraints and observed connectivity, reducing the size of the network.
- **Scenario reduction:** The number of demand scenarios is limited in stochastic models to maintain tractability.

These design choices ensure that the optimization model remains solvable within a reasonable computational time while preserving the essential structure of the logistics system.

6.4 Results

6.4.1 Network Reduction and Tractability

The original logistics network contains 103 centers and over 5,000 potential connections, with distances ranging from 12 km to 5,164 km. Such a dense network is computationally intractable for multi-scenario optimization.
After applying service feasibility and data-driven filtering, the network is reduced to 25 candidate arcs, connecting 16 origin centers, 9 intermediate centers, and 58 demand nodes.
Interestingly, the resulting network is dominated by direct shipments (CDC $\rightarrow$ demand nodes), with no CDC $\rightarrow$ RDC arcs retained after filtering. This indicates that, for the selected data subset, historical operations favor direct or near-direct routing.
This reduction significantly decreases model complexity while preserving operationally relevant connections, enabling efficient optimization across multiple approaches.

6.4.2 Baseline Performance: Expected Value Model

The Expected Value (EV) model yields an average total cost of 3686.82 and an average shortage of 35 units. A key structural observation is that the EV solution relies entirely on direct shipments, with no transshipment flows. This is a direct consequence of using average demand, which underestimates peak demand and eliminates the need for flexible routing.
As a result, the EV model experiences the highest shortage among all approaches, demonstrating the limitations of deterministic planning under demand uncertainty.

6.4.3 Impact of Scenario-Based Optimization

The Sample Average Approximation (SAA) model improves performance over EV, reducing average cost to 3533.58 and average shortage to 34.5.

This improvement is driven by incorporating demand variability across multiple scenarios. However, given the limited number of scenarios (S = 2) (due to the limited memory of the Gorobi license), the model does not fully capture the range of demand uncertainty.

As a result, while SAA provides a more robust solution than EV, its performance remains relatively close to the deterministic baseline.

6.4.4 Impact of Predictive Modeling

The Predict-Then-Optimize (PTO) approach significantly improves performance. The Decision Tree-based PTO model achieves a total cost of 1402.95 with a shortage of 13 units, while the kNN-based PTO model yields a slightly higher cost of 1451.15 and a shortage of 13.42 units.

These results demonstrate that incorporating demand predictions substantially improves decision quality. The Decision Tree model outperforms kNN because it can capture structured patterns and partition demand more effectively. However, both PTO approaches treat predicted demand as certain, limiting their ability to hedge against uncertainty.

6.4.5 Predictive–Prescriptive Optimization Results

The Predictive–Prescriptive (PP) models achieve the best performance across all approaches. The kNN-based PP model yields the lowest cost of 282.84 and a shortage of 2.67 units, while the Decision Tree-based PP model achieves a cost of 450.00 and a shortage of 4.5 units.

This represents a dramatic improvement over both PTO and SAA approaches.

The superior performance of PP models arises from their ability to incorporate uncertainty through data-driven scenario weighting. Unlike PTO, which relies on a single prediction, PP models consider multiple relevant demand realizations, enabling more flexible and informed decisions.

The kNN-based PP model performs best because it captures local demand variability, whereas the Decision Tree produces coarser scenario groupings.

6.4.6 Summary of Results

The overall summaries of all methods are given below:

Table 1: Summary of Model Performance

Method	Avg Total Cost	Avg Shortage
EV	3686.82	35.00
SAA	3533.58	34.50
PTO (Decision Tree)	1402.95	13.00
PTO (kNN)	1451.15	13.42
PP (Decision Tree)	450.00	4.50
PP (kNN)	282.84	2.67

6.5 Evaluation

6.5.1 Cost Performance Comparison

Result: Predictive–Prescriptive models significantly outperform all other approaches in terms of total cost, with PP (kNN) achieving the lowest average cost (See Figure 1).

Interpretation: The figure shows a clear performance hierarchy. EV and SAA models incur the highest costs,

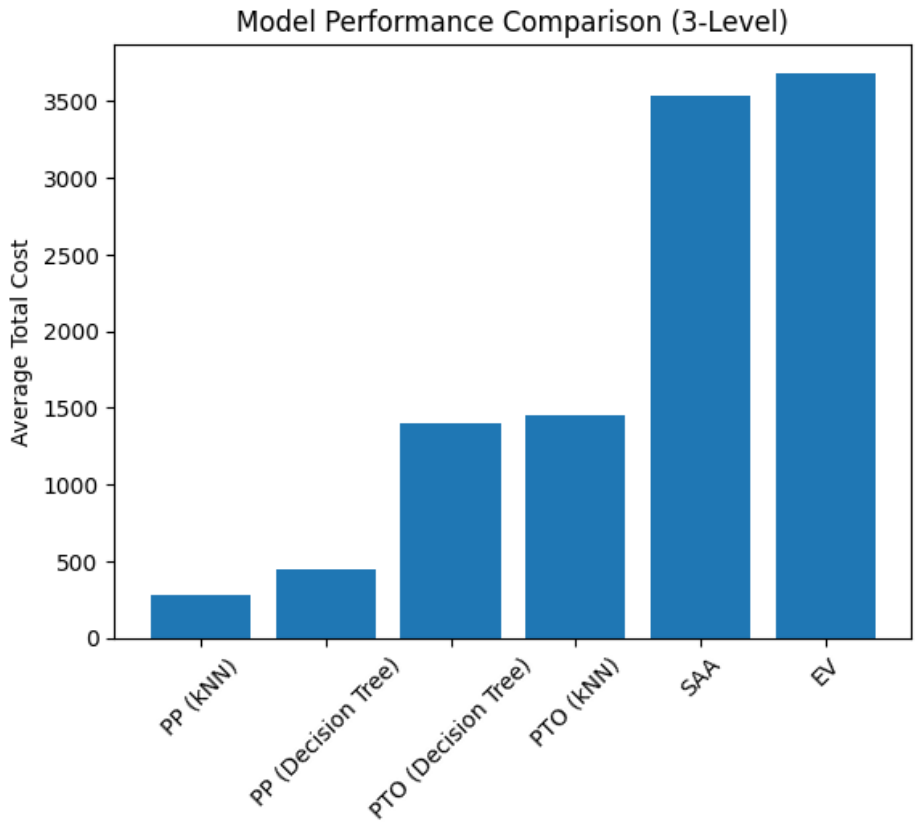

Figure 1: Average Total Cost Across Models

while PTO models reduce costs substantially. The PP models achieve the most significant reduction, with PP (kNN) performing best.

Underlying Reason: This pattern arises from how each method handles uncertainty. EV ignores uncertainty entirely, leading to under-prepared decisions. SAA incorporates uncertainty but treats all scenarios equally, limiting its effectiveness. PTO improves performance using predictions but ignores residual uncertainty. PP models outperform all others because they integrate prediction and optimization, allowing decisions to adapt to locally relevant demand patterns.

6.5.2 Relative Cost Reduction vs SAA

Result: PTO models provide modest improvements over SAA, while PP models achieve the largest cost reductions.

Table 2: Relative Cost Reduction Compared to SAA

Model	Relative Cost Reduction vs SAA
PP (kNN)	0.9200
PP (Decision Tree)	0.8727
PTO (Decision Tree)	0.6030
PTO (kNN)	0.5893
SAA	0.0000
EV	-0.0434

Interpretation: The figure highlights that predictive methods outperform purely statistical approaches. However, the most significant gains are achieved when prediction is integrated into the optimization process. **Underlying Reason:** SAA treats all scenarios equally, whereas PTO uses a single predicted scenario. PP models go further by assigning weights to scenarios based on similarity, allowing the model to focus on the most relevant demand patterns. This leads to more efficient allocation decisions and greater cost reduction.

6.5.3 Prediction Error Analysis

Result: The prediction model achieves a mean absolute error (MAE) of 0.7 and RMSE of 1.23.
Interpretation: These error values indicate reasonably accurate predictions at the aggregated level. However, even small prediction errors can propagate into significant operational inefficiencies when used directly in optimization.
Underlying Reason: The PTO framework relies entirely on point predictions, which do not capture uncertainty. As a result, even accurate predictions may lead to suboptimal decisions under variability, highlighting the limitation of prediction-only approaches.

6.5.4 Decision Error Analysis

Result: All models exhibit large decision errors, with average absolute errors exceeding 4,300 units.
Interpretation: This indicates a significant mismatch between planned deliveries and realized demand, even for advanced models.
Underlying Reason: The high decision error reflects the inherent uncertainty in demand and the limitations of the network structure. Even with predictive or stochastic methods, constraints such as limited supply, restricted arcs, and capacity limits prevent perfect demand matching.

6.5.5 Risk Analysis

Result: Predictive–Prescriptive models significantly reduce unmet demand compared to all other approaches, with PP (kNN) achieving the lowest average shortage (See Figure 2).
Interpretation: The figure shows a clear reduction in unmet demand as we move from EV and SAA models toward predictive and prescriptive approaches. EV and SAA exhibit the highest shortage levels, while PTO models substantially reduce shortage. The PP models achieve the lowest shortage, indicating superior service performance.
Underlying Reason: This pattern reflects how each method handles demand uncertainty. EV ignores variability entirely, leading to systematic underestimation of demand. SAA incorporates uncertainty but lacks contextual weighting, limiting its effectiveness. PTO improves performance using predictions but remains vulnerable to prediction errors. PP models outperform all others by incorporating multiple relevant demand scenarios through data-driven weighting, allowing the system to better hedge against uncertainty and reduce unmet demand.
Managerial Insight: Reducing shortage requires not only accurate prediction but also uncertainty-aware decision-making. Predictive–Prescriptive approaches provide a more reliable framework for improving service levels in uncertain environments.

6.6 Sensitivity Analysis: Shortage Penalty

Result: The total cost increases with the shortage penalty, while unmet demand and transportation costs remain largely unchanged (See Figure 3).
Interpretation: The system is insensitive to changes in penalty levels for operational decisions. Increasing the penalty does not reduce unmet demand.
Underlying Reason: This indicates that the model is operating under structural constraints, such as limited supply or restricted network feasibility. Since these constraints prevent additional demand fulfillment, increasing penalties only inflates total cost without improving service levels.

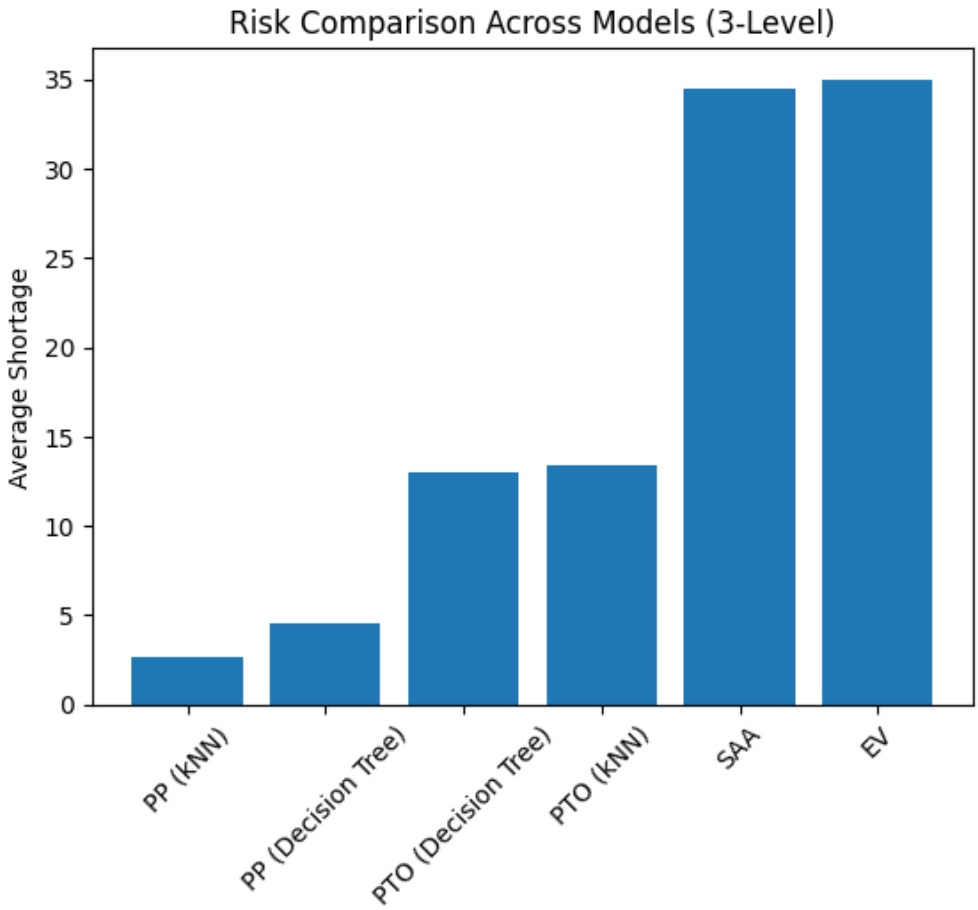

Figure 2: Risk Comparison Across Models (Average Shortage)

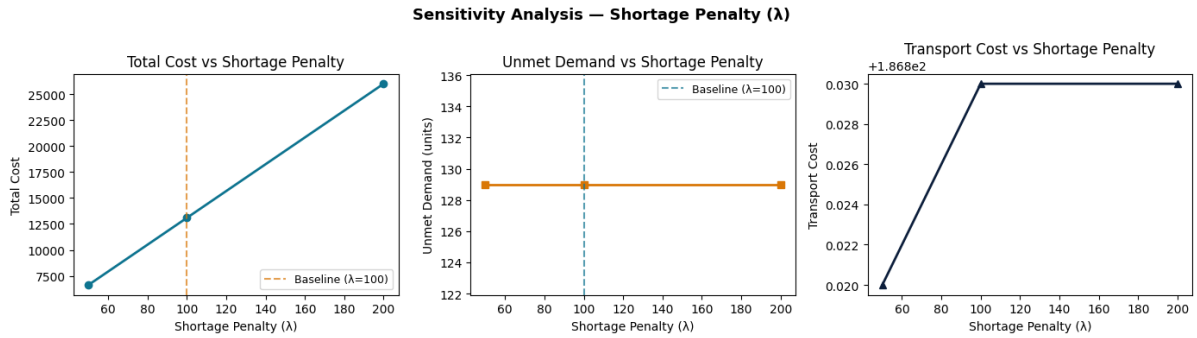

Figure 3: Sensitivity Analysis with Respect to Shortage Penalty

7 Managerial Insights

The results of this study provide several important managerial insights for logistics planning under demand uncertainty.

i) Integrating prediction with optimization is critical. The Predictive–Prescriptive (PP) framework consistently outperforms all alternative approaches in both cost and service performance. While predictive models alone (PTO) significantly improve decision quality, they remain vulnerable to uncertainty. By contrast, PP models incorporate uncertainty directly into the optimization process, enabling more robust and adaptive decisions. This highlights the importance of moving beyond traditional forecast-then-optimize pipelines toward fully integrated data-driven decision frameworks.

ii) Ignoring uncertainty leads to poor operational performance. The Expected Value (EV) model exhibits the highest cost and shortage levels, demonstrating that deterministic planning is inadequate in uncertain environments. Even the SAA approach, while incorporating scenarios, provides only marginal improvements due to limited scenario diversity. This suggests that simplistic representations of uncertainty can lead to systematically underprepared logistics plans.

iii) Local demand patterns matter more than global averages. The superior performance of the PP (kNN) model indicates that capturing local demand variability is essential. Unlike Decision Trees, which create coarse partitions, kNN leverages similarity-based learning to identify relevant historical patterns. This allows the system to respond more effectively to context-specific demand fluctuations.

iv) Service improvement requires structural changes, not just parameter tuning. The sensitivity analysis shows that increasing shortage penalties does not reduce unmet demand, indicating that the system is constrained by supply, capacity, or network structure. This implies that improving service levels requires operational changes such as increasing capacity, expanding feasible routes, or redesigning the network, rather than simply adjusting cost parameters.

v) Direct shipping dominates in simplified networks. The network reduction results reveal that most optimal solutions rely on direct CDC-to-demand shipments, with minimal use of transshipment centers. While this improves computational tractability, it may limit flexibility. Managers should carefully evaluate whether additional transshipment capacity could improve robustness in more complex or larger-scale settings.

vi) A trade-off between model complexity and scalability is unavoidable. The aggregation and arc-filtering strategies significantly reduce computational complexity, enabling tractable optimization. However, this simplification may omit important routing options. Managers must balance model realism with computational feasibility when deploying such systems in practice.

Improving service performance requires structural changes (e.g., increased capacity or expanded network connec-

tivity) rather than parameter tuning.

8 Recommendations

Based on the findings, we propose the following recommendations for implementing data-driven logistics planning systems:

i) Adopt Predictive–Prescriptive frameworks: Organizations should prioritize integrated approaches that combine machine learning with optimization, as they provide substantial improvements in both cost efficiency and service reliability.

ii) Invest in high-quality demand data and feature engineering: The effectiveness of PP models depends heavily on the quality of input data and the ability to capture relevant demand patterns. Enhancing data collection and feature design will directly improve decision outcomes.

iii) Expand network flexibility: Since structural constraints limit performance, firms should consider increasing routing options, relaxing service-time constraints where feasible, or expanding capacity at key nodes.

iv) Use predictive models cautiously: While PTO models improve performance, they can lead to overconfident decisions if prediction errors are ignored. Incorporating uncertainty remains essential.

v) Implement scenario-based decision support tools: Even when full PP integration is not feasible, using weighted or context-aware scenarios can improve planning quality compared to traditional methods.

9 Limitations

Despite its contributions, this study has several limitations:

i) Limited data size: The dataset contains a relatively small number of observations, which restricts the ability to fully capture demand variability and limits the effectiveness of scenario-based methods.

ii) Simplified network structure: The aggregation from a four-level to a three-level network improves tractability but may omit important operational details, particularly in last-mile delivery.

iii) Approximate supply estimation: Supply is inferred from historical shipments rather than observed inventory or capacity data, introducing potential inaccuracies.

iv) Restricted scenario set: The number of scenarios in SAA and PP models is limited due to computational constraints, which may affect robustness.

v) Static planning framework: The model operates on a weekly planning horizon and does not account for dynamic or real-time decision adjustments.

10 Future Extensions

Several directions can be explored to enhance this framework:

i) Dynamic and real-time optimization: Extending the model to incorporate rolling-horizon or real-time decision-making would improve responsiveness to demand changes.

ii) Full multi-echelon network modeling: Future work can incorporate the full four-level network structure, including detailed last-mile operations, to improve realism.

iii) Advanced machine learning models: More sophisticated models such as gradient boosting, deep learning, or probabilistic forecasting methods could further improve demand representation.

iv) Robust and risk-averse optimization: Incorporating risk measures such as CVaR or worst-case optimization could provide stronger guarantees under extreme demand scenarios.

v) Scalability enhancements: Decomposition techniques, parallel computing, or heuristic methods can be explored to scale the approach to large real-world networks.

vi) Incorporating operational constraints: Future models could include vehicle routing, scheduling, and work-

force constraints to better reflect real logistics systems.

11 Conclusion

This study develops a data-driven prescriptive planning framework for logistics networks operating under demand uncertainty and evaluates multiple approaches for integrating predictive models with optimization. Using a three-level aggregated representation of the RRS logistics network, we compare deterministic, stochastic, predictive, and predictive-prescriptive methods within a unified framework.

The results demonstrate a clear performance hierarchy across approaches. Deterministic methods such as the Expected Value (EV) model perform poorly because they cannot capture demand variability, leading to high costs and significant unmet demand. Stochastic methods such as Sample Average Approximation (SAA) provide only limited improvements, as their effectiveness depends heavily on the number and representativeness of scenarios. Predict-Then-Optimize (PTO) approaches significantly improve performance by leveraging machine learning predictions but remain vulnerable to prediction errors because they lack explicit uncertainty modeling.

The Predictive-Prescriptive (PP) framework achieves the best performance, delivering substantial reductions in both total cost and unmet demand. In particular, the PP (kNN) model outperforms all other methods by capturing local demand variability and incorporating it directly into the optimization process through data-driven scenario weighting.

These findings strongly align with the challenges highlighted in the RiRiShun logistics dataset study Guo et al. (2024), which emphasizes the complexity of real-world logistics systems and the critical role of demand uncertainty in operational decision-making. Our results provide empirical evidence that traditional deterministic and even basic stochastic approaches are insufficient for such environments, and that more advanced data-driven frameworks are necessary.

In addition, the sensitivity analysis reveals that system performance is often constrained by structural factors such as network design and capacity limitations, rather than cost parameters alone. This observation reinforces the practical insights from the RiRiShun study, which found that operational feasibility and network structure play central roles in determining system performance.

Overall, this work contributes to the growing literature on predictive-prescriptive analytics by demonstrating its practical value in large-scale logistics planning. The proposed framework provides a scalable, flexible approach that can be extended to more complex supply chain systems, offering significant potential to improve operational efficiency and service reliability in real-world applications.

References

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